

Meme Stock Predictor: Complete Tech Stack with Justifications

1. Development Environment

IDE & Tools

Technology	Justification
VS Code	Lightweight with excellent Python support, Jupyter integration, and AI plugins like GitHub Copilot for rapid prototyping.
Jupyter Lab	Essential for exploratory data analysis (EDA) of stock/sentiment data. Supports visualization libraries like Plotly/Matplotlib.

Experiment Tracking

Technology	Justification
Weights & Biases (W&B)	Tracks hyperparameters, model versions, and metrics across experiments. Critical for comparing meme pattern detection models.
DVC (Data Version Control)	Version-controls datasets (Reddit/Twitter scrapes) alongside code to ensure reproducibility.

2. Data Pipeline

Data Collection

Technology	Justification
snscraper	No API rate limits for Twitter/Reddit scraping. Bypasses restrictions vs. official APIs.
Apache NiFi	Orchestrates real-time data flows from multiple sources (e.g., Reddit → Kafka → S3).

Storage

Technology	Justification
Delta Lake	ACID transactions for meme stock data. Handles schema evolution when new social platforms are added.
TimescaleDB	Optimized for time-series data (1-min stock prices, sentiment scores). Enables fast window queries.

3. Machine Learning Stack

NLP & Sentiment

Technology	Justification
FinBERT	Pre-trained for financial text but fine-tuned on meme slang (e.g., "moon" → +0.8 sentiment).
Emoji2Vec	Custom embeddings for 🚀/💎 emojis. Captures hype intensity better than text alone.

Time-Series

Technology	Justification
Prophet	Handles holiday effects (e.g., "Elon tweets on weekends"). Better for crypto than ARIMA.
PyTorch Lightning	Simplifies LSTM training on GPU for pump pattern detection. Supports gradient clipping for stability.

4. Production Deployment

Model Serving

Technology	Justification
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Triton Inference Server	Supports ensemble models (e.g., NLP + time-series). GPU-optimized for <50ms latency.
FastAPI	Async support for high-throughput alert APIs. Auto-generates OpenAPI docs for subscribers.

Monitoring

Technology	Justification
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Prometheus	Tracks API latency, model drift (e.g., "Twitter sentiment accuracy drops below 60%").
Evidently AI	Detects feature drift when meme slang evolves (e.g., new terms like "skrrt").

5. Monetization Stack

API Monetization

Technology	Justification
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Stripe	Handles subscription tiers (e.g., \$29/mo for basic alerts, \$299/mo for API access).
Kafka	Streams real-time predictions to paying hedge funds. Persistent queues prevent data loss.

White-Labeling

Technology	Justification
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Next.js	Allows custom branding for crypto exchange partners. SSR improves SEO for signal pages.

Key Architectural Diagrams

Data Flow

```mermaid

flowchart LR

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A[Twitter/Reddit] --> B{Data Lake}
B --> C[Feature Store]
C --> D((ML Model))
D --> E[API]
E --> F[Alerts]
E --> G[Partner Dashboards]
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